

Emotional Changes in Japanese Blog Space Resulting from the 3.11 Earthquake

Yukie Sano, Hideki Takayasu, and Misako Takayasu

Abstract We quantified the emotional changes observed in social media after major disasters, focusing especially on the Japanese blog space after the Great East Japan Earthquake in 2011. We checked the appearances of Japanese adjectives and found that special emotion adjectives, such as ‘impatient’, and ‘frustrating’ which involve the want to help others but the person has no means to and feels frustrated, occur with considerably increasing frequency. To visualize social mood, we drew a co-occurrence network of adjectives showing a major topological change at the site of the quake. Measuring emotional changes after an emergency has been difficult, but, our research has the potential to achieve it.

1 Introduction

Social data collected on a large scale has helped to promote our understanding of society and ourselves during this century [1–3]. Drastic changes in information technology enable us to collect various types of communication data, such as those on mobile phone and face-to-face contacts. Hidden patterns of human activities are uncovered, and these results are beginning to be used to solve real social problems.

Web-based data attract many scientists particularly because it is easy to collect data on a large scale for discussion in scientific papers and such data reflect various real social phenomena such as political, market, financial, and daily events. Examples include Facebook—the world’s largest social networking service [4–6]—

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Google and Yahoo! search engines [7–9], Wikipedia—a multilingual online encyclopaedia [10–14]—YouTube—an online movie site [15]—and Twitter—a microblogging platform on which users can post messages up to 140 characters [16–22].

Since various large amounts of social media data are available, studies regarding social media have been greatly increasing substantially. In particular, new applications are demanding and attracted, many researches have focused on the connections between real-world phenomena and social media. For example, Mandel et al. focused on the emotional changes in 65,000 tweets during half the month when hurricane Irene approached the U.S. and found that the level of concern had increased depending on region and gender [23]. However, little is known about long-term emotional changes in social media.

There are many ways to detect emotions in texts. Some examples are Linguistic Inquiry and Word Count (LIWC) [24], Positive and Negative Affect Schedule (PANAS) [25], Affective Norms for English Words (ANEW) [26], and Point Of Mood State (POMS) [27, 28]. Furthermore, Emoticons (combination of ‘emotion’ and ‘icon’) such as ‘:)’ and WordNet a thesaurus have been used to detect emotions [29–32]. However, it is known that resulting polarity (i.e., positive or negative) is different depending on the methods [33]. Additionally, most of the methods are English, therefore, there is a not strict dictionary to detect emotions from Japanese texts.

In this paper, we use more than 3.2 billion Japanese blog posts for six years since 1 November 2007 as typical Japanese texts. Here we simply use whole Japanese adjectives to detect emotions. Our observation periods include the Great East Japan earthquake in 2011 which is said to have changed social mood qualitatively. First, we describe our data and method in Section 2. Next we compare the relative frequencies of adjectives before and after the quake and draw co-occurrence networks for the adjectives to visualize social emotions in Section 3. Summary and discussion are in Section 4.

2 Data and Methods

2.1 Data

We studied Japanese blog data from 1 November 2007 to 31 October 2013 (2192 days) using an Internet service called ‘Kuchikomi@kakaricho¹’ to collect data. This fee-charging service provides an original web-site and an application programming interface (API). The API returns the daily number of blogs in which any given target word appears in a given period. The daily number of blogs containing a blog post which includes that the target word occurring more than once is counted. Thus, if one blog post includes a target word multiple times, the API counts it as one.

¹ <http://kakaricho.jp> (Accessed: 11 March 2015)

The API also provides a spam filter with three levels—weak, middle and strong—depending on the desired spam detection accuracy. Here, we use the middle level of spam filter to collect data. The full API database contains more than 3.2 billion blog posts from 38 million accounts.

We search for adjectives listed in the MeCab²—a Japanese morphological analyzer—dictionary with original surface forms using the API. Because there are various ways of conjugating forms in Japanese, we search for just their original form for simplicity; there are 1741 adjectives in total. Subsequently, we summarize these adjectives' time series in the case that they have the same pronunciations and meanings. Because Japanese uses three different character sets—Hiragana, Katakana and Kanji (Chinese characters)—instead of an alphabet, people tend to use words with different surface forms that have the same pronunciations and meanings. This procedure leads to 839 adjectives. Finally, we removed extremely low- (less than 10 times per day) or high-frequency (more than 100,000 times per day) adjectives, resulting in a total of 550 adjectives' time series.

2.2 *z-test for the quality of two proportions*

In this research, we apply a *z*-test for the quality of two proportions to determine whether adjective *i*'s occurrence $x_i(t)$ differs at different times T_0 and T_1 . The *z*-score of statistic z_i is calculated as follows:

$$z_i = \frac{\left| \frac{x_i(T_0)}{X(T_0)} - \frac{x_i(T_1)}{X(T_1)} \right|}{\sqrt{X_i(1-X_i) \left[\frac{1}{x_i(T_0)} + \frac{1}{x_i(T_1)} \right]}} \quad (1)$$

where $X(t)$ is the total number of blogs at time t and X_i is calculated as follows:

$$X_i = \frac{x_i(T_0) + x_i(T_1)}{X(T_0) + X(T_1)} \quad (2)$$

The null hypothesis is that z_i follows a standard normal distribution. Therefore, we can calculate the *p*-value to check the proportions of adjectives in different time periods.

2.3 *Partial correlation coefficient*

To visualize the change in social emotions at the quake, we construct adjective co-occurrence networks for the pre- and post- quake periods. First, we calculate the Pearson's linear correlation coefficient r_{ij} between the frequency $y_i(t)$ and $y_j(t)$ of

² <https://code.google.com/p/mecab/> (Accessed: 11 March 2015)

adjectives i and j respectively, with a difference of normalized time series,

$$y_i(t) = \frac{x_i(t+1)}{X(t+1)} - \frac{x_i(t)}{X(t)}. \quad (3)$$

Pearson's linear correlation coefficient r_{ij} between adjectives i and j is calculated as follows:

$$r_{ij} = \frac{\sum_t (y_i(t) - \langle y_i \rangle) (y_j(t) - \langle y_j \rangle)}{\sigma_i \sigma_j} \quad (4)$$

where σ_i and σ_j are the standard deviations of $y_i(t)$ and $y_j(t)$, respectively. Here, we use three weeks before and after the quake for the comparison, with $T = 21$ data points. If adjectives i and j have a positive correlation at the 0.01 significant level, then the value of r_{ij} is greater than 0.5487 for $T = 21$.

To extract more essential links from the co-occurrence network, we use the partial correlation coefficient r_{ij}^k as follows:

$$r_{ij}^k = \frac{r_{ij} - r_{ik} \cdot r_{jk}}{\sqrt{1 - r_{ik}^2} \sqrt{1 - r_{jk}^2}}. \quad (5)$$

r_{ij}^k is the partial correlation between i and j under the fixed condition of k . Partial correlation is equivalent to the correlation between residuals $y_i(t)$ and $y_j(t)$ after the removing correlation between each r_{ik} and r_{jk} .

Normally, we can calculate the partial correlation D_{ij} that is removed by all other adjectives' effects by calculating an inverse correlation matrix. However, we cannot calculate it that way in this case, because we have only $T = 21$ data points for each time series and there are 550 adjectives (samples). Therefore, we calculate the partial correlation D_{ij} for each pair of adjectives i and j by removing k 's effect and checking the following condition.

$$D_{ij} = \begin{cases} \max_k r_{ij}^k & (\text{if } \forall k, r_{ij}^k \geq D_c) \\ 0 & (\text{otherwise}). \end{cases} \quad (6)$$

where $D_c = 0.5613$ at the 0.01 significance level of partial correlation. To stress the significant partial correlation coefficient, we use maximum value of r_{ij}^k in this research.

3 Results

3.1 Adjectives in emergency and normal periods

We focus on relative changes in adjectives before and after the quake by calculating z_i , as shown in Eq. (1). Here, we define the pre-quake period T_0 as three weeks

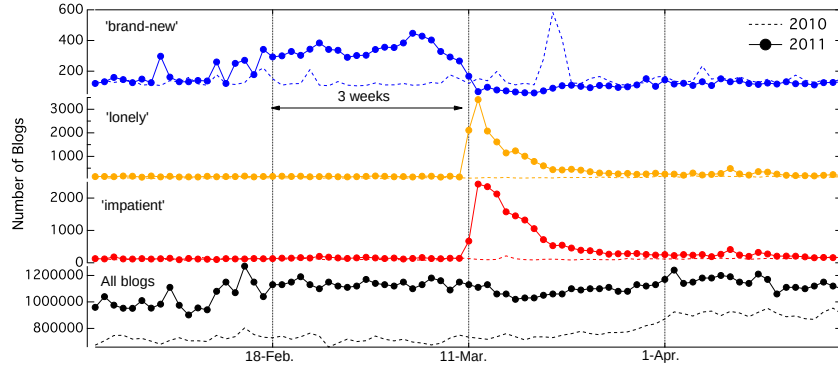


Fig. 1 Daily number of blogs including ‘brand-new’, ‘lonely’, and ‘impatient’ from the top. Daily number of whole blogs $X(t)$ is shown in the bottom. Solid lines are in 2011 and dashed lines are in 2010 for comparison. Note that sudden increase around 20 March 2010 for ‘brand-new’ is caused by spam blogs, because we confirm that it diminished when we search with the strong level of spam filter.

before the quake, from 18 February to 10 March in 2011. Similarly, we define the post-quake period T_1 as three weeks after the quake, from March 12 to April 1 in 2011.

According to the results of the calculation of z_i for all 550 possible adjectives, 72 adjectives increased and 74 adjectives decreased significantly at the 0.01 significant level. Tables 1 and 2 show top 10 increased and decreased adjectives respectively. Figure 1 shows examples of increased and decreased adjectives.

We found that adjectives such as ‘impatient’ which express users’ feelings of frustration have increased considerably according to Table 1 (#1,4,5,6). Even these adjectives have different surface forms, albeit similar meanings, e.g. the frustrated feeling of wanting to help others but being unable to do so.

The usage of words such as ‘heartless’ and ‘shameless’ have also increased significantly, according to Table 1 (#7,8,10). Some people behaved selfishly, buying food and bottles of water under despite the severe shortage conditions after the quake. Therefore, blog posts included complaints about these behaviors with the increased use of these adjectives.

On the other hand, ‘brand-new’, ‘earlier’, and ‘cannot wait’ in Table 2 (#1,2,5) decreased significantly. These words are often used for positive meanings in expectation of new seasons such as spring and goods such as movies. In fact, many companies canceled or postponed their releases of new product and events. For example, the ceremony of Kyushu Shinkansen opening was canceled, iPad2 (a popular tablet

Table 1 Adjectives that increased significantly ($p < 0.01$) after the quake.

#	Word	Original form (Japanese)	$x_i(T_1)/x_i(T_0)$
1	Impatient	はがゆい, 歯がゆい, 歯痒い	5.91
2	Lonely	こころぼそい, 心細い	5.48
3	Precious	とうとい, 貴い, 尊い	4.84
4	Sorry	こころぐるしい, 心苦しい	4.36
5	Frustrating	もどかしい	4.13
6	Cannot stand the situation	いたたまれない	4.07
7	Heartless	こころない, 心無い	3.57
8	Shameless	あさましい, 浅ましい	3.50
9	Barely afford	ままならない	3.42
10	Base	さかしい	3.40

Table 2 Adjectives that decreased significantly ($p < 0.01$) after the quake.

#	Word	Original form (Japanese)	$x_i(T_1)/x_i(T_0)$
1	Brand new	めあたらしい, 目新しい	0.29
2	A little earlier	ひと足早い	0.31
3	Itchy	かゆい, 痒い	0.52
4	Tawdry	やすっぽい, 安っぽい	0.56
5	Cannot wait	まちどおしい, 待ち遠しい, 待遠しい	0.58
6	Salty-sweet	あまからい, 甘辛い	0.59
7	Celebration	めでたい, メデタイ, 目出度い	0.62
8	Bitter	しぶい, 渋い	0.63
9	Spicy	こうばしい, 香ばしい, 芳ばしい	0.63
10	Tough	てごわい, 手ごわい	0.63

device) release was postponed³, and many scheduled movie releases were canceled or postponed⁴. As a result, people lost many chances to use such words.

Furthermore, we found that adjectives related to taste such as ‘salty-sweet’ and ‘bitter’ decreased significantly according to Table 2 (#6,8,9). These decreases of words may reflect the so-called ‘self-restraint mood’ that people stop to have parties outside such as annual cherry blossom viewing party and to go restaurant. Consequently, the words regarding taste, e.g. restaurant and cook reviews, could decrease. These decreased adjectives seem to be more related to social activities rather than emotions, compared to increased adjectives. Therefore, estimating economic situations by using the adjectives is the crucially interesting future topic.

³ http://gigazine.net/news/20110316_apple-delays-ipad2-launch/ (Accessed: 11 March 2015)

⁴ <http://www.cinematoday.jp/page/N0031042> (Accessed: 11 March 2015)

3.2 Adjective changes in the co-occurrence network

We constructed co-occurrence network of adjectives during the pre-quake period T_0 and post-quake period T_1 . The node size indicates the relative frequency of words compared with the entire period and the color corresponds to the community it belongs to. The community is decided by the modularity Q as follows:

$$Q = \frac{1}{2M} \sum_{i,j=1}^N \left(A_{ij} - \frac{k_i k_j}{2M} \right) \delta(c_i, c_j), \quad (7)$$

where $N = 550$ and M are total number of nodes and links in network respectively. A_{ij} is the weight of the link between node i and j . Here the weight is correlation coefficient r_{ij} calculated in Eq. (4). $k_i = \sum_j A_{ij}$ is the sum of the weights linked to node i , and c_i is the community which i belongs to. $\delta(c_i, c_j)$ is 1 if i and j belong to same community ($c_i = c_j$), otherwise 0. In this paper, we maximize Q for undirected weighted network by a software named Gephi⁵ (version 0.8.2) to detect community with the algorithm introduced by Blondel et al. [34].

There are 3354 links among 550 nodes during T_0 using the normal correlation coefficient r_{ij} (Fig. 2 (Left)) and 61 links using the partial correlation coefficient D_{ij} (Fig. 2 (Right)). As expected, the node size is nearly the same because there was no major news before the quake T_0 and there are no special properties in these networks. There are 14 communities in the network. The largest community shares 16.36% and the second shares 14.55%.

In contrast, there are two major communities post-quake period T_1 . There are 6125 links among the 550 nodes in T_1 by calculating r_{ij} (Fig. 3 (Top)) and 73 links are found by calculating the partial correlation coefficient (Fig. 3 (Bottom)). There are 5 communities in the network. The largest community shares 26.57% and the second shares 21.59%. Thus, more nodes categorized into the same communities than T_0 period. This confirms that one community corresponds to the adjectives that increased significantly as shown in the previous section.

4 Summary and Discussion

We observed social emotions in the Japanese blog space during emergency periods, especially before and after the Great East Japan Earthquake. Here, we focus on the relative changes of adjectives. We found that the feelings such as wanting to help others but having no means and the feeling of frustration increased considerably. Thus, adjectives such as ‘impatient’, ‘sorry’, and ‘frustrating’ increased.

To visualize the change in social mood during the quake, we constructed adjective co-occurrence networks. We found that there is a clear topological difference between the pre- and post-quake periods. We found one large community in the post-

⁵ <http://gephi.org> (Accessed: 11 March 2015)

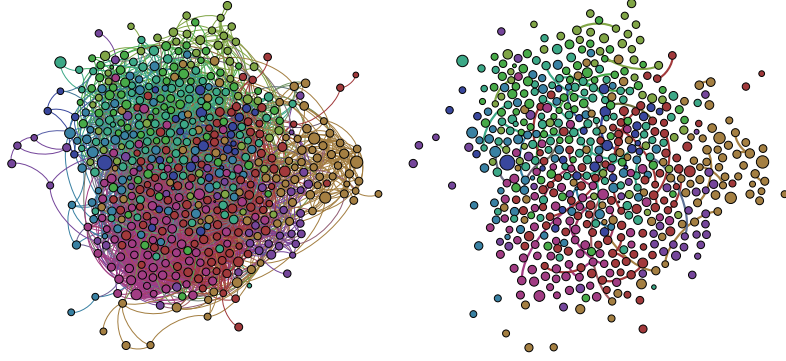


Fig. 2 Correlation networks consisting of adjectives before the quake T_0 . (Left) Links are drawn on the basis of the correlation coefficient from Eq. (4). (Right) Links are drawn on the basis of the partial correlation coefficient from Eq. (6). Nodes are colored by their community and sized by their relative appearances $\langle x_i(T_0) \rangle / \langle x_i \rangle$, where $\langle x_i \rangle$ is the mean in the entire period and $\langle x_j(T_0) \rangle$ is the mean in the period T_0 .

quake networks with increased adjectives. This result suggests that people tended to share similar emotions post-quake period.

Our results were derived from a limited case study of Japanese social media during the 3.11 Earthquake. However, our results still have novelty and potential, because no one could record small messages from normal people accurately during emergencies until the advent of social media.

Given that during the quake, rumors and false information are said to have spread via mobile phones and social media [35], investigating social emotions during emergency periods has the potential to provide useful warnings regarding false information diffusion. Because psychologists have observed, on the basis of experiments with limited numbers of subjects, that rumors are more likely to diffuse in an emergency situation [36] and when people feel anxious [37].

We expect that data assimilation—a combination of agent-based simulation and real data analysis—will assist us in preventing potential secondary disasters such as false information diffusion and riots during emergencies. The Internet era has even said to foster ‘digital wildfires in a hyperconnected world [38]’ similar to the spread of real wildfires. Our research sheds light on this universal problem and could issue warning signals of potential digital wildfires. We hope that our results can be applied to prevent such problems in the future.

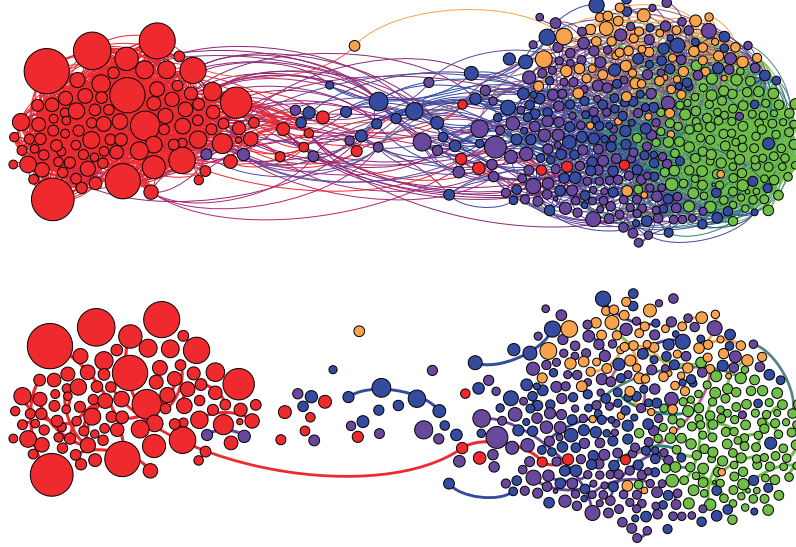


Fig. 3 Correlation networks consisting of adjectives after the quake T_1 . (Top) Links are drawn on the basis of the correlation coefficient from Eq. (4). (Bottom) Links are drawn on the basis of the partial correlation coefficient from Eq. (6). There is one large grouping that consists of increased frequency nodes soon after the quake (red). Nodes are colored by their community and sized by their relative appearances $\langle x_i(T_1) \rangle / \langle x_i \rangle$, where $\langle x_i(T_1) \rangle$ is the mean in the T_1 period .

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1 Introduction

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Google and Yahoo! search engines [7–9], Wikipedia—a multilingual online encyclopaedia [10–14]—YouTube—an online movie site [15]—and **Twitter—a microblogging platform on which users can post messages up to 140 characters** [16–22].

Since various large amounts of social media data are available, studies regarding social media have been greatly increasing substantially. **Twitter is a microblogging platform on which users can post messages of up to 140 characters. Twitter reflects political [16,17,18] and financial [19,20] activities, major events and news [21,22,23] and everyday life [24,25].**

Moreover, people use Twitter and other social media during emergencies [26,27,28,29,30,31]. Of course, people in critical situations cannot use social media, but other people use them to search for missing persons, detect dangerous area and blocked street, and provide information regarding necessary relief supplies.

On the other hand, false information is said to diffuse significantly [32]. Because rapid application use is needed to rescue people, research has focused primarily on the connections between real-world phenomena and social media. **In particular, new applications are demanding and attracted, many researches have focused on the connections between real-world phenomena and social media. For example,** Mandel et al. focused on the emotional changes in 65,000 tweets during half the month when hurricane Irene approached the U.S. and found that the level of concern had increased depending on region and gender [23]. However, little is known about long-term emotional changes in social media. **Here we focus on the emotional changes during the Great East Japan Earthquake (the 3.11 Earthquake) drawing on a larger number Japanese blogs for a longer period. In particular, we quantify the length of the emergency period by determining emotional texts in the blog space.**

There are many ways to detect emotions in texts. Linguistic Inquiry and Word Count (LIWC), People express positive feelings in tweets on weekends according LIWC [24]. Similarly Positive and Negative Affect Schedule (PANAS) [34], categorizes emotions into 20 dimensions. Affective Norms for English Words (ANEW) is a tool for determining three emotional dimensions—valence, arousal, dominance—by rating 1034 words on a scale of 1–9. The sadness expressed in Twitter space at the sudden death of Michael Jackson was studied using ANEW [35]. Furthermore, Amazon Mechanical Turk (AMT) and machine learning methods can determine emotions by scoring arbitrary words [36]. Emoticons (combination of ‘emotion’ and ‘icon’) such as ‘:)’ and WordNet a thesaurus have been used to detect emotions [25,37,38]. Bollen et al. predicted with an accuracy of 86.7% the daily increase and decrease changes in the closing values of the DJIA using GPOMS, a combination of Point Of Mood State (POMS) [39], a questionnaire investigation to measure examinees’ psychological state, and Google search word tokens [19].

Some examples are Linguistic Inquiry and Word Count (LIWC) [24], Positive and Negative Affect Schedule (PANAS) [25], Affective Norms for English Words (ANEW) [26], and Point Of Mood State (POMS) [27, 28]. Furthermore, Emoticons (combination of ‘emotion’ and ‘icon’) such as ‘:)’ and WordNet a thesaurus have been used to detect emotions [29–32]. However, it is known that resulting polarity (i.e., positive or negative) is different depending on the methods [33]. Additionally,

most of the methods are English, therefore, there is a not strict dictionary to detect emotions from Japanese texts.

In this paper, we use ~~adjectives to detect the dynamical changes in social emotions from~~ more than 3.2 billion Japanese blog posts for six years since 1 November 2007 ~~as typical Japanese texts. Here we simply use whole Japanese adjectives to detect emotions.~~ Our observation periods include the Great East Japan earthquake in 2011 which is said to have changed social mood qualitatively. First, we describe our data and method in Section 2. Next we compare the relative frequencies of adjectives before and after the quake and draw co-occurrence networks for the adjectives to visualize social emotions in Section 3. ~~Summary and discussion are in Section 4.~~

2 Data and Methods

2.1 Data

We studied Japanese blog data from 1 November 2007 to 31 October 2013 (2192 days) using an Internet service called ‘Kuchikomi@kakaricho¹’ to collect data. This fee-charging service provides an original web-site and an application programming interface (API). The API returns the daily number of blogs in which any given target word appears in a given period. The daily number of blogs containing a blog post which includes that the target word occurring more than once is counted. Thus, if one blog post includes a target word multiple times, the API counts it as one. The API also provides a spam filter with three levels—weak, middle and strong—depending on the desired spam detection accuracy. Here, we use the middle level of spam filter to collect data. The full API database contains more than 3.2 billion blog posts from 38 million accounts.

We search for adjectives listed in the MeCab²—a Japanese morphological analyzer—dictionary with original surface forms using the API. Because there are various ways of conjugating forms in Japanese, we search for just their original form for simplicity; there are 1741 adjectives in total. Subsequently, we summarize these adjectives’ time series in the case that they have the same pronunciations and meanings. Because Japanese uses three different character sets—Hiragana, Katakana and Kanji (Chinese characters)—instead of an alphabet, people tend to use words with different surface forms that have the same pronunciations and meanings. This procedure leads to 839 adjectives. Finally, we removed extremely low- (less than 10 times per day) or high-frequency (more than 100,000 times per day) adjectives, resulting in a total of 550 adjectives’ time series.

¹ <http://kakaricho.jp> (Accessed: 11 March 2015)

² <https://code.google.com/p/mecab/> (Accessed: 11 March 2015)

2.2 *z-test for the quality of two proportions*

In this research, we apply a *z*-test for the quality of two proportions to determine whether adjective *i*'s occurrence $x_i(t)$ differs at different times T_0 and T_1 . The *z*-score of statistic z_i is calculated as follows:

$$z_i = \frac{\left| \frac{x_i(T_0)}{X(T_0)} - \frac{x_i(T_1)}{X(T_1)} \right|}{\sqrt{X_i(1-X_i) \left[\frac{1}{x_i(T_0)} + \frac{1}{x_i(T_1)} \right]}} \quad (1)$$

where $X(t)$ is the total number of blogs at time t and X_i is calculated as follows:

$$X_i = \frac{x_i(T_0) + x_i(T_1)}{X(T_0) + X(T_1)} \quad (2)$$

The null hypothesis is that z_i follows a standard normal distribution. Therefore, we can calculate the *p*-value to check the proportions of adjectives in different time periods.

2.3 *Partial correlation coefficient*

To visualize the change in social emotions at the quake, we construct adjective co-occurrence networks for the pre- and post- quake periods. First, we calculate the Pearson's linear correlation coefficient r_{ij} between the frequency $y_i(t)$ and $y_j(t)$ of adjectives *i* and *j* respectively, with a difference of normalized time series,

$$y_i(t) = \frac{x_i(t+1)}{X(t+1)} - \frac{x_i(t)}{X(t)}. \quad (3)$$

Pearson's linear correlation coefficient r_{ij} between adjectives *i* and *j* is calculated as follows:

$$r_{ij} = \frac{\sum_t (y_i(t) - \langle y_i \rangle) (y_j(t) - \langle y_j \rangle)}{\sigma_i \sigma_j} \quad (4)$$

where σ_i and σ_j are the standard deviations of $y_i(t)$ and $y_j(t)$, respectively. Here, we use three weeks before and after the quake for the comparison, with $T = 21$ data points. If adjectives *i* and *j* have a positive correlation at the 0.01 significant level, then the value of r_{ij} is greater than 0.5487 for $T = 21$.

To extract more essential links from the co-occurrence network, we use the partial correlation coefficient r_{ij}^k as follows:

$$r_{ij}^k = \frac{r_{ij} - r_{ik} \cdot r_{jk}}{\sqrt{1 - r_{ik}^2} \sqrt{1 - r_{jk}^2}}. \quad (5)$$

r_{ij}^k is the partial correlation between i and j under the fixed condition of k . Partial correlation is equivalent to the correlation between residuals $y_i(t)$ and $y_j(t)$ after the removing correlation between each r_{ik} and r_{jk} .

Normally, we can calculate the partial correlation D_{ij} that is removed by all other adjectives' effects by calculating an inverse correlation matrix. However, we cannot calculate it that way in this case, because we have only $T = 21$ data points for each time series and there are 550 adjectives (samples). Therefore, we calculate the partial correlation D_{ij} for each pair of adjectives i and j by removing k 's effect and checking the following condition.

$$D_{ij} = \begin{cases} \max r_{ij}^k & (\text{if } \forall k, r_{ij}^k \geq D_c) \\ 0 & (\text{otherwise}). \end{cases} \quad (6)$$

where $D_c = 0.5613$ at the 0.01 significance level of partial correlation. To stress the significant partial correlation coefficient, we use maximum value of r_{ij}^k in this research.

3 Results

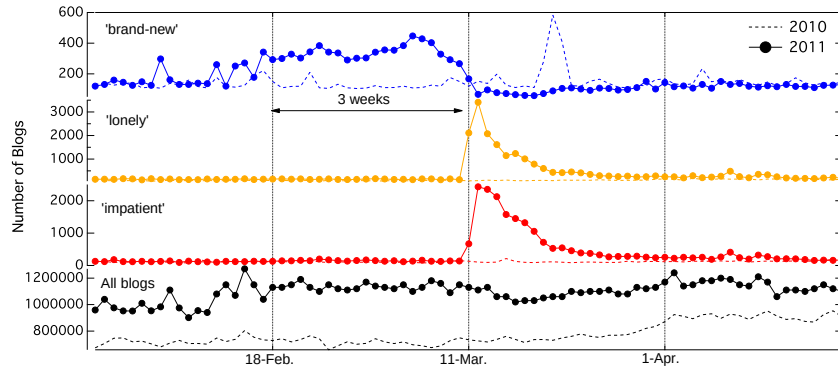


Fig. 1 Daily number of blogs including ‘brand-new’, ‘lonely’, and ‘impatient’ from the top. Daily number of whole blogs $X(t)$ is shown in the bottom. Solid lines are in 2011 and dashed lines are in 2010 for comparison. Note that sudden increase around 20 March 2010 for ‘brand-new’ is caused by spam blogs, because we confirm that it diminished when we search with the strong level of spam filter.

Table 1 Adjectives that increased significantly ($p < 0.01$) after the quake.

#	Word	Original form (Japanese)	$x_i(T_1)/x_i(T_0)$
1	Impatient	はがゆい, 歯がゆい, 歯痒い	5.91
2	Lonely	こころぼそい, 心細い	5.48
3	Precious	とうとい, 貴い, 尊い	4.84
4	Sorry	こころぐるしい, 心苦しい	4.36
5	Frustrating	もどかしい	4.13
6	Cannot stand the situation	いたたまれない	4.07
7	Heartless	こころない, 心無い	3.57
8	Shameless	あさましい, 浅ましい	3.50
9	Barely afford	ままならない	3.42
10	Base	さもない	3.40

3.1 Adjectives in emergency and normal periods

We focus on relative changes in adjectives before and after the quake by calculating z_i , as shown in Eq. (1). Here, we define the pre-quake period T_0 as three weeks before the quake, from 18 February to 10 March in 2011. Similarly, we define the post-quake period T_1 as three weeks after the quake, from March 12 to April 1 in 2011.

According to the results of the calculation of z_i for all 550 possible adjectives, 72 adjectives increased and 74 adjectives decreased significantly at the 0.01 significant level. Tables 1 and 2 show top 10 increased and decreased adjectives respectively. Figure 1 shows examples of increased and decreased adjectives.

We found that adjectives such as ‘impatient’ which express users’ feelings of frustration have increased considerably according to Table 1 (#1,4,5,6). Even these adjectives have different surface forms, albeit similar meanings, e.g. the frustrated feeling of wanting to help others but being unable to do so.

The usage of words such as ‘heartless’ and ‘shameless’ have also increased significantly, according to Table 1 (#7,8,10). Some people behaved selfishly, buying food and bottles of water under despite the severe shortage conditions after the quake. Therefore, blog posts included complaints about these behaviors with the increased use of these adjectives.

On the other hand, ‘brand-new’, ‘earlier’, and ‘cannot wait’ in Table 2 (#1,2,5) decreased significantly. These words are often used for positive meanings in expectation of new seasons such as spring and goods such as movies. In fact, many companies canceled or postponed their releases of new product and events. For example, the ceremony of Kyushu Shinkansen opening was canceled, iPad2 (a popular tablet device) release was postponed³, and many scheduled movie releases were canceled or postponed⁴. As a result, people lost many chances to use such words.

³ <http://gigazine.net/news/20110316-apple-delays-ipad2-launch/> (Accessed: 11 March 2015)

⁴ <http://www.cinematoday.jp/page/N0031042> (Accessed: 11 March 2015)

Table 2 Adjectives that decreased significantly ($p < 0.01$) after the quake.

#	Word	Original form (Japanese)	$x_i(T_1)/x_i(T_0)$
1	Brand new	めあたらしい, 目新しい	0.29
2	A little earlier	ひと足早い	0.31
3	Itchy	かゆい, 痒い	0.52
4	Tawdry	やすっぽい, 安っぽい	0.56
5	Cannot wait	まちどおしい, 待ち遠しい, 待遠しい	0.58
6	Salty-sweet	あまからい, 甘辛い	0.59
7	Celebration	めでたい, メデタイ, 目出度い	0.62
8	Bitter	しぶい, 渋い	0.63
9	Spicy	こうばしい, 香ばしい, 芳ばしい	0.63
10	Tough	てごわい, 手ごわい	0.63

Furthermore, we found that adjectives related to taste such as ‘salty-sweet’ and ‘bitter’ decreased significantly according to Table 2 (#6,8,9). These decreases of words may reflect the so-called ‘self-restraint mood’ that people stop to have parties outside such as annual cherry blossom viewing party and to go restaurant. Consequently, the words regarding taste, e.g. restaurant and cook reviews, could decrease. These decreased adjectives seem to be more related to social activities rather than emotions, compared to increased adjectives. Therefore, estimating economic situations by using the adjectives is the crucially interesting future topic.

This result suggests that people tend to stop posting blogs regarding including happy events, e.g. food or restaurant reviews. This result indicates that many blog posts included the same adjectives, reflecting a situation in which people shared the same emotions after the quake.

3.2 Adjective changes in the co-occurrence network

We constructed co-occurrence network of adjectives during the pre-quake period T_0 and post-quake period T_1 . The node size indicates the relative frequency of words compared with the entire period and the color corresponds to the community it belongs to. The community is decided by the modularity Q as follows:

$$Q = \frac{1}{2M} \sum_{i,j=1}^N \left(A_{ij} - \frac{k_i k_j}{2M} \right) \delta(c_i, c_j), \quad (7)$$

where $N = 550$ and M are total number of nodes and links in network respectively. A_{ij} is the weight of the link between node i and j . Here the weight is correlation coefficient r_{ij} calculated in Eq. (4). $k_i = \sum_j A_{ij}$ is the sum of the weights linked to node i , and c_i is the community which i belongs to. $\delta(c_i, c_j)$ is 1 if i and j belong to same community ($c_i = c_j$), otherwise 0. In this paper, we maximize Q for undirected

weighted network by a software named Gephi⁵ (version 0.8.2) to detect community with the algorithm introduced by Blondel et al. [34].

There are 3354 links among 550 nodes during T_0 using the normal correlation coefficient r_{ij} calculated from Eq. (4) (Fig. 2 (Left)) and 61 links using the partial correlation coefficient D_{ij} (Fig. 2 (Right)). We calculated the partial correlation coefficients using Eq. (6) to extract more essential links, resulting in 61 links in the partial correlation network in Fig. 2 (Right). As expected, the node size is nearly the same because there was no major news before the quake T_0 and there are no special properties in these networks. There are 14 communities in the network. The largest community shares 16.36% and the second shares 14.55%.

In contrast, there are two major groupings communities post-quake period T_1 . There are 6125 links among the 550 nodes after the quake period in T_1 by calculating r_{ij} the correlation coefficient in Eq. (4) (Fig. 3 (Top)) and 73 links are found by calculating the partial correlation coefficient in Eq. (6) (Fig. 3 (Bottom)). There are 5 communities in the network. The largest community shares 26.57% and the second shares 21.59%. Thus, more nodes categorized into the same communities than T_0 period. This confirms that one community corresponds to the adjectives that increased significantly as shown in the previous section.

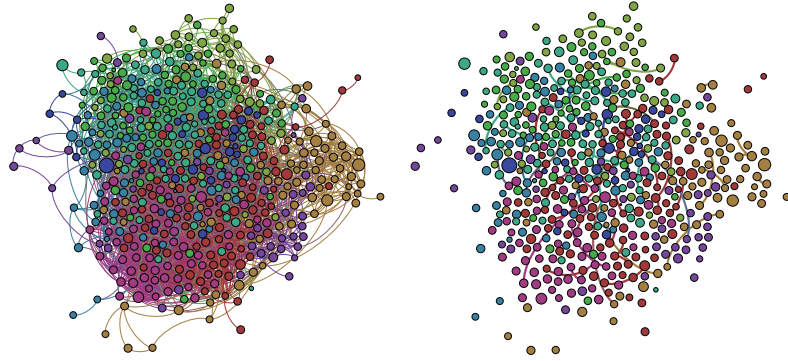


Fig. 2 Correlation networks consisting of adjectives before the quake T_0 . (Left) Links are drawn on the basis of the correlation coefficient from Eq. (4). (Right) Links are drawn on the basis of the partial correlation coefficient from Eq. (6). Nodes are colored by their community and sized by their relative appearances $\langle x_i(T_0) \rangle / \langle x_i \rangle$, where $\langle x_i \rangle$ is the mean in the entire period and $\langle x_i(T_0) \rangle$ is the mean in the period T_0 .

⁵ <http://gephi.org> (Accessed: 11 March 2015)

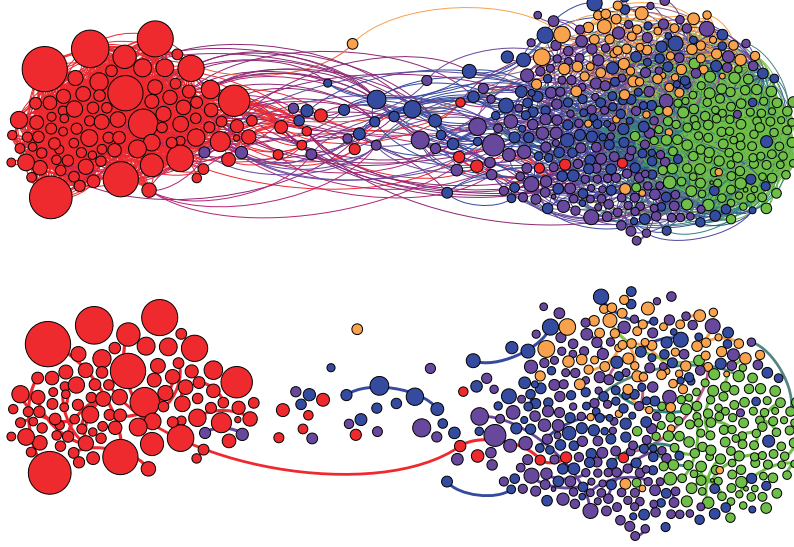


Fig. 3 Correlation networks consisting of adjectives after the quake T_1 . (Top) Links are drawn on the basis of the correlation coefficient from Eq. (4). (Bottom) Links are drawn on the basis of the partial correlation coefficient from Eq. (6). There is one large grouping that consists of increased frequency nodes soon after the quake (red). Nodes are colored by their community and sized by their relative appearances $\langle x_i(T_1) \rangle / \langle x_i \rangle$, where $\langle x_i(T_1) \rangle$ is the mean in the T_1 period .

4 Summary and Discussion

We observed social emotions in the Japanese blog space during emergency periods, especially before and after the Great East Japan Earthquake. Here, we focus on the relative changes of adjectives. We found that the feelings such as wanting to help others but having no means and the feeling of frustration increased considerably. Thus, adjectives such as ‘impatient’, ‘sorry’, and ‘frustrating’ increased.

To visualize the change in social mood during the quake, we constructed adjective co-occurrence networks. We found that there is a clear topological difference between the pre- and post-quake periods. We found one large community in the post-quake networks with increased adjectives. This result suggests that people tended to share similar emotions post-quake period.

Our results were derived from a limited case study of Japanese social media during the 3.11 Earthquake. However, our results still have novelty and potential, because no one could record small messages from normal people accurately during emergencies until the advent of social media.

Given that during the quake, rumors and false information are said to have spread via mobile phones and social media [35], investigating social emotions during emer-

gency periods has the potential to provide useful warnings regarding false information diffusion. Because psychologists have observed, on the basis of experiments with limited numbers of subjects, that rumors are more likely to diffuse in an emergency situation [36] and when people feel anxious [37].

We expect that data assimilation—a combination of agent-based simulation and real data analysis—will assist us in preventing potential secondary disasters such as false information diffusion and riots during emergencies. ~~In particular, false information diffusion during emergencies has occurred frequently since ancient Roman times.~~ The Internet era has even said to foster ‘digital wildfires in a hyperconnected world [38]’ similar to the spread of real wildfires. Our research sheds light on this universal problem and could issue warning signals of potential digital wildfires. We hope that our results can be applied to prevent such problems in the future.

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